Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

An aqueous solution of a salt X turns blood red on treatment with CNS^- and blue on treatment with $K_4[Fe(CN)_6]$. X also gives a positive chromyl chloride test. The salt X is :

(A) $CuCl_2$

(B) $FeCl_3$

(C) $Cu(NO_3)_2$

(D) $Fe(NO_3)_3$

Q.2

For detection of sulphur in an organic compound sodium nitroprusside is added to the sodium extract. A violet colour is obtained due to formation of

(A) $Fe(CN)_2$

(B) $K_3Fe(CN)_5NS$

(C) $Na_4[Fe(CN)_5NOS]$

(D) $Na_4Fe(CN)_6$

Q.3

The cation that will not be precipitated by H_2S in the presence of dil. HCl is :

(A) Pb^{2+}

(B) Cu^{2+}

(C) Co^{2+}

(D) As^{3+}

Q.4

0.5 g mixture of $K_2Cr_2O_7$ and $KMnO_4$ was treated with excess of KI in acidic medium. I_2 liberated required $100\ cm^3$ of $0.15\ N Na_2S_2O_3$ solution for titration. The percentage amount of $K_2Cr_2O_7$ in the mixture is

(A) 85.36%

(B) 14.64%

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(C) 58.63%

(D) 26.14%

Q.5

A white sodium salt dissolves readily in water to give a solution which is neutral to litmus. When silver nitrate solution is added to the solution, a white precipitate is obtained which does not dissolve in dil. HNO_3. The anion could be

(A) CO_3^{2-}

(B) Cl^-

(C) SO_4^{2-}

(D) S^{2-}

Q.6

An organic compound has C and H percentage in the ratio $6:1$ and C and O percentage in the ratio $3:4$ the compound is

(A) $HCHO$

(B) CH_3OH

(C) CH_3CH_2OH

(D) $(COOH)_2$

Q.7

A salt which gives CO_2 with hot conc. H_2SO_4 and also decolourizes acidified $KMnO_4$ on warming is:

(A) HCO_3^-

(B) CO_3^{2-}

(C) Oxalate

(D) Acetate

Q.8

The sodium extract prepared from sulphanilic acid, contains SCN^-. It gives blood red colouration with

(A) a mixture of Na_2 S and CS_2

(B) $FeCl_3$

(C) $FeSO_4$

(D) Na_2SO_3

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Q.9

An orange precipitate of II group is dissolved in conc. HCl; the solution when treated with excess of water turns milky due to formation of

(A) $Sn(OH)Cl$

(B) $Sb(OH)Cl_2$

(C) $SbOCl$

(D) $Sb(OH)_2Cl$

Q.10

A substance on treatment with dil.H $_2SO_4$ liberates a colourless gas which produces
(I) turbidity with baryta water and
(II) turns acidified dichromate solution green. The reaction indicates the presence of

(A) CO_3^{2-}

(B) S^{2-}

(C) SO_3^{2-}

(D) NO_3^-

Q.11

An aqueous solution of $FeSO_4$, $Al_2(SO_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The materials obtained are:-

(A) A colourless filtrate and a green residue.

(B) A yellow filtrate and a green residue.

(C) A yellow filtrate and a brown residue.

(D) A green filtrate and a brown residue.

Q.12

Which of the following statements is incorrect?

(A) Fe^{2+} ion also gives blood red colour with SCN^- ion.

(B) Fe^{3+} ion also gives blood red colour with SCN^- ion.

(C) On passing H_2S into Na_2ZnO_2 solution a white ppt of ZnS is formed.

(D) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[Cu(NH_3)_4]^{2+}$ ion.

Q.13

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What is the red coloured solid compound that is insoluble in water which becomes soluble if some KI is added to water and also heating the solid in a test tube results in liberation of some violet coloured fumes and droplets of a metal appear on the cooler parts of the test tube?

(A) $(NH_4)_2Cr_2O_7$

(B) HgI_2

(C) HgO

(D) Pb_3O_4

Q.14

SO_2 and CO_2 gas can be differentiated by

(A) MnO_4^-

(B) $Cr_2\,O_7^{2-}$

(C) lime water

(D) Both A & B

Q.15

A fire work gave bright crimson light. It probably contain an element of

(A) Ca

(B) Sr

(C) Ba

(D) Mg

Answers & Solutions

Q.1 Answer:

$FeCl_3$

Solution:

Fe^{3+}, radical gives blood red with CNS^-

$Fe^{3+} + 3CNS^- \longrightarrow Fe(CNS)_3$

Ferric sulphocyanide (Blood red colouration) Fe^{3+} gives blue colour on treatment with $K_4\,[Fe(CN)_6]$

$4Fe^{3+} + 3\,K_4\,[Fe(CN)_6] \longrightarrow$

$12\,K^+ + Fe_4\,[Fe(CN)_6]_3$

Ferricferrocyanide (Prussian blue) Cl^- radical gives chromyl chloride test.

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Q.2 Answer:

$Na_4\,[Fe(CN)_5NOS]$

Solution:

$$Na_2S + Na_2[Fe(CN)_5\,NO] \longrightarrow$$

Sodium nitroprusside

$$Na_4[Fe(CN)_5NOS]$$

Sodium thionitroprusside

Q.3 Answer:

Co^{2+}

Solution:

Co^{+2} **ion is precipitated by** $H_2\,S$ **in presence of** NH_4OH **which is a group reagent of group IV in cationic analysis.**

Q.4 Answer:

14.64%

Solution:

Let the amount of the $K_2Cr_2O_7$ **in the mixture be** x **g, then amount of** $KMnO_4$ **will be** $(0.5 - x)g \therefore \quad \left(\frac{x}{49} + \frac{0.5-x}{31.6}\right) = \frac{100\times0.15}{1000}$

where 49 is Eq. wt. of $K_2Cr_2O_7$ **and 31.6 is**

Eq. wt. of $KMnO_4$**. On solving, we get** $x = 0.073$ g

Percentage of $K_2Cr_2O_7$

$= \frac{0.0732\times100}{0.5} = 14.64\%$

Q.5 Answer:

Cl^-

Solution:

$NaCl$ **is a salt of strong acid and strong base hence on dissolution will give neutral solution.**

$NaCl + AgNO_3 \longrightarrow AgCl\downarrow + HNO_3$

Q.6 Answer:

$HCHO$

Solution:

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% ratio of C : H is 6 : 1 and C : O is 3 : 4 or 6 : 8 $\therefore$ % ratio of C : H : O is 6 : 1 : 0

(Total $= 15$)

% of C $= \frac{6}{15} \times 100 = 40$

$40/12 = 3.33 = 1$

% of H $= \frac{1}{15} \times 100 = 6.6$

$6.6/1 = 6.6 = 2$

% of O $= \frac{8}{15} \times 100 = 53.3$

$53.3/16 = 3.3 = 1$

simple ratio: CH_2O

$\therefore$ The compound is $HCHO$

Q.7 Answer:

Oxalate

Solution:

$$Na_2C_2O_4 + H_2SO_4 \rightarrow$$

$$Na_2SO_4 + H_2O + CO + CO_2$$

CO burns with blue flame $2CO + O_2 \rightarrow 2CO_2$

$$Na_2C_2O_4 + H_2SO_4(dil) \rightarrow$$

$$Na_2SO_4 + H_2C_2O_4$$

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \rightarrow$$

$$\rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$$

Q.8 Answer:

$FeCl_3$

Solution:

$$SCN^- + Fe^{3+} \longrightarrow Fe(SCN)_3(bloodredppt)$$

Q.9 Answer:

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$SbOCl$

Solution:

$$Sb_2S_3 + 6HCl \rightarrow 2SbCl_3 + 3H_2S$$

$$SbCl_3 + 6H_2O \rightarrow \underset{\text{ppt (White)}}{SbOCl} + 2HCl$$

Q.10 Answer:

SO_3^{2-}

Solution:

$$Na_2SO_3 + 2HCl(dil) \rightarrow 2NaCl + H_2O + SO_2$$

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow \underset{\text{Green}}{K_2SO_4 + Cr_2(SO_4)_3} + H_2O$$

Q.11 Answer:

A yellow filtrate and a green residue.

Solution:

Green residue is due to $Fe(OH)_2$ and yellow filtrate is due to chromate ions CrO_4^{2-}

Q.12 Answer:

Fe^{2+} ion also gives blood red colour with SCN^- ion.

Solution:

Only Fe^{3+} ions give blood red colouration with SCN^- ions.

$$Fe^{3+} + SCN^- \longrightarrow [Fe(SCN)]^{2+}$$

(dark red)

Q.13 Answer:

HgI_2

Solution:

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HgI_2

Q.14 Answer:

Both A & B

Solution:

..

Q.15 Answer:

Sr

Solution:

Ca = **Brick red**

Sr = **red/crimson**

Ba = **Apple green**

Mg = **no flame test**

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